

Formal Specification Document (v2.0) Not a derivation notebook nor  
implementation guide

# ETAMv2.0: Energetic–Temporal–Environmental Activity Model, Version 2.0

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## 1 Notation and Conventions

All symbols are defined as follows unless otherwise noted:

- **A–T–E** (Spatial-Temporal-Energetic): primitive ontological basis **A** for *spatial-architectural activity*, **T** for *temporal-slot activation*, **E** for *energo-material response*
- **XR** (ATE Basis Encoding): discrete coordinate representation of the **A–T–E** (Spatial-Temporal-Energetic) basis;  
 $R = [\mathbf{n}_1, \dots, \mathbf{n}_N]^\top \in \mathbb{Z}^{N \times 3}$
- **icR** (Complex Generator:  $f(R) = x(R) + i c(R)$ ): complex-valued generator  $f(R) = x(R) + i c(R)$ , where  $x$  is the *covariant observable*,  $c$  the *contravariant lifting potential*

## 2 Introduction

The **Energetic–Temporal–Environmental Activity Model (ETAM)** is a physics-informed, differentiable framework for predicting macroscopic scalar outputs  $Y$  from discrete-continuous hybrid inputs. Version 2.0 introduces unified slot-based encoding, complex-valued response functions, and adaptive multilevel hierarchy driven by the imaginary component.

**Clarification of novelty:** The **A–T–E** (Spatial-Temporal-Energetic) triad is not a re-labeling of conventional physical quantities. It is a *primitive ontological basis* defined by its role in the **icR** (Complex Generator:  $f(R) = x(R) + i c(R)$ ) generator: **A** encodes spatial-architectural constraints, **T** encodes temporal-slot resolution, **E** encodes energo-material coupling. No existing model uses this triad as a unified coordinate system.

This aligns with Rosen’s relational ontology: **A–T–E** (Spatial-Temporal-Energetic) does not describe *what things are*, but *how they co-constitute observables through constrained coordination* ?.

## 3 Core Output Structure

The model defines a dual-domain mapping:

$$Y = A \cdot T \cdot E \tag{1}$$

$$f(R) = x(R) + i c(R) \tag{2}$$

where  $Y \in \mathbb{R}_{\geq 0}$  is the physical observable (e.g., binding energy in eV), and  $f(R) \in \mathbb{C}$  is the **icR** (Complex Generator:  $f(R) = x(R) + i c(R)$ ) generator. Alignment requires  $\Re[f(R)] \approx Y$ , while  $\Im[f(R)] = c(R) \geq 0$  governs scale-level selection.

## 4 Input Representation $R$

Each active unit  $j$  (atom or functional group) is encoded as a discrete triple:

$$\mathbf{n}_j = (n_{A,j}, n_{T,j}, n_{E,j}) \in \mathbb{Z}_{\geq 0}^3. \tag{3}$$

For  $N$  units, the input tensor is:

$$R = [\mathbf{n}_1, \mathbf{n}_2, \dots, \mathbf{n}_N]^\top \in \mathbb{Z}^{N \times 3}. \quad (4)$$

$R$  is not a feature vector it is the *coordinate representation of the A–T–E (Spatial-Temporal-Energetic) basis in discrete state space*. Each row  $\mathbf{n}_j$  is a point in the A–T–E (Spatial-Temporal-Energetic) lattice, where  $n_{A,j}, n_{T,j}, n_{E,j}$  are *mutually constraining coordinates*, not independent scalars.

## 5 Component Definitions

### 5.1 A-value: Spatial-Architectural Activity

Let  $\omega(p) \in \{0, 1, 2\}$  denote occupancy (empty/single electron/lone pair) at orbital slot  $p$ . Define slot weights  $k(p)$ :

$$k(p) = \begin{cases} 1 & \text{for } s\text{-orbitals,} \\ 2 & \text{for } p\text{-orbitals,} \\ 3 & \text{for } d\text{-orbitals,} \\ 4 & \text{for } f\text{-orbitals.} \end{cases} \quad (5)$$

The activity constant for unit  $j$  is:

$$C_{A,j} = \sum_{p \in \{s,p,d,f\}} \omega(p) k(p). \quad (6)$$

Then:

$$A_j = \alpha_A \cdot C_{A,j}, \quad A = \frac{1}{N} \sum_{j=1}^N A_j, \quad (7)$$

with scaling factor  $\alpha_A > 0$ .

### 5.2 T-value: Temporal-Slot Activation

Six discrete time slots  $k = 0, \dots, 5$  cover  $1 \times 10^{-15}$  s to  $1 \times 10^6$  s. Let  $\nu(k) \in \{0, 1, 2\}$  indicate activation level. Then:

$$C_{T,j} = \sum_{k=0}^5 \nu(k) w(k), \quad T_j = \alpha_T \cdot C_{T,j}, \quad T = \frac{1}{N} \sum_{j=1}^N T_j. \quad (8)$$

### 5.3 E-value: Energo-Material Response Field

Four environmental response slots: ionization ( $e_I$ ), affinity ( $e_A$ ), polarization ( $e_P$ ), solvation ( $e_S$ ). Let  $\mu(e) \in \{0, 1, 2\}$ . Define:

$$C_{E,j} = \sum_{e \in \{I,A,P,S\}} \mu(e) w_e. \quad (9)$$

Environment-modulated field:

$$\Phi_I(\Xi) = \frac{F^2}{\mathbb{I}E}, \quad (10)$$

$$\Phi_P(\Xi) = \frac{\epsilon - 1}{\epsilon + 2} F^2, \quad (11)$$

$$\Phi_S(\Xi) = \frac{\Delta G_{\text{solv}}}{RT}, \quad (12)$$

and:

$$E_j = \alpha_E \cdot C_{E,j} \cdot \left[ 1 + \sum_e \underbrace{\frac{\mu(e) w_e}{C_{E,j}}}_{\beta_e(C_{E,j})} \cdot \Phi_e(\Xi) \right], \quad E = \frac{1}{N} \sum_{j=1}^N E_j. \quad (13)$$

## 6 Response Function $f(R)$

The **icR** (Complex Generator:  $f(R) = x(R) + i c(R)$ ) framework ( $f(R) = x(R) + i c(R)$ ) is not an engineering choice for numerical stability. It is a *structural necessity*: only complex-valued representation permits simultaneous encoding of covariant observables ( $x$ ) and contravariant scale selectors ( $c$ ) within a single differentiable map.

The icR mapping realizes a *Connes-type spectral geometry*?: discrete **A–T–E** (Spatial-Temporal-Energetic) coordinates  $R$  define a finite spectral triple, whose complex extension  $f(R)$  yields a differentiable continuum limit.

### 6.1 Real Part (Covariant Observable)

$$x(R) = \alpha_x \cdot \langle C_A \rangle \cdot \langle C_T \rangle \cdot \langle C_E \rangle, \quad (14)$$

or alternatively via MLP:

$$x(R) = \text{MLP}_{\theta_x}(\log(C_A + \varepsilon), \log(C_T + \varepsilon), \log(C_E + \varepsilon)). \quad (15)$$

### 6.2 Imaginary Part (Contravariant Lifting Potential)

$$c(R) = \lambda \cdot \left[ \frac{\text{Var}(C_A)}{\langle C_A \rangle^2} + \frac{\text{Var}(C_T)}{\langle C_T \rangle^2} \right] \cdot (1 + \eta \cdot S_{\text{env}}), \quad (16)$$

where  $S_{\text{env}} = \sum_e \Phi_e(\Xi)$ . The non-negative imaginary part  $c(R)$  implements Penrose’s ‘ladder of infinity’ on the complex plane?: each threshold  $\tau_k$  marks a rung where the system’s effective description ascends to a new ontological level.

- $c(R) < \tau_1$ : Level 1 (molecular determinism)
- $\tau_1 \leq c(R) < \tau_2$ : Level 2 (mesoscale clusters)
- $c(R) \geq \tau_2$ : Level 3 (condensed-phase statistical fields)

## 7 Training Objective

$$\mathcal{L} = \|\Re[f(R)] - Y\|^2 + \gamma \cdot \mathcal{L}_{\text{reg}}(c(R)), \quad (17)$$

where  $\mathcal{L}_{\text{reg}}$  enforces level consistency (e.g., entropy penalty on  $c(R)$ ) or other domain-informed constraints.

## 8 Constrained Inverse Queries

The model supports *constrained inverse queries*: given a partial configuration  $\tilde{R}$  and target scalar  $y_{\text{target}}$ , the set of completions  $R_{\text{miss}}$  satisfying both energy fidelity and level consistency is non-empty and computable under standard differentiable optimization.

This inverse capability is not an add-on feature. It is a *logical consequence* of the **A–T–E** (Spatial-Temporal-Energetic)–**icR** (Complex Generator:  $f(R) = x(R) + i c(R)$ ) duality: if  $f$  is bijective over its domain (up to level constraints), then generation is the algebraic inverse of prediction?.

## 9 Version Identification

**ETAMv2.0** is formally defined by this specification (v2.0, 2026-02-22). Key normative innovations:

1. Unified discrete slot encoding across **A**, **T**, and **E** as a primitive ontological basis;
2. Complex-valued separation of covariant observable ( $x$ ) and contravariant lifting potential ( $c$ );
3. Imaginary part-driven adaptive multilevel hierarchy grounded in Penrose’s ladder;
4. Verifiable scientific discovery via Wigner-style bidirectional invertibility.

## References

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Eugene Wigner. The unreasonable effectiveness of mathematics in the natural sciences. 1960.